

## Midterm Exam Solutions:

### 1. (25 points, 5 points for each part)

Suppose we have three events  $A, B, C$  with  $P(A) = 0.3$ ,  $P(B) = 0.4$ , and  $P(C) = 0.5$ . Suppose we also know that  $P(A \cap B) = 0.1$ ,  $P(A \cap C) = .2$ ,  $P(B \cap C) = 0.3$ , and  $P(A \cap B \cap C) = .01$ . Find the probability of each of the following events:

- (a)  $A \cup B$
  - (b)  $A^c \cap B^c$
  - (c)  $A \cup B \cup C$
  - (d)  $A^c \cap B^c \cap C^c$
  - (e)  $(A^c \cap B^c) \cup C$
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$$\begin{aligned} a) \quad P(A \cup B) &= P(A) + P(B) - P(A \cap B) \quad \text{by inclusion-exclusion principle} \\ &= 0.3 + 0.4 - 0.1 \end{aligned}$$

$$P(A \cup B) = 0.6$$

$$\begin{aligned} b) \quad P(A^c \cap B^c) &= 1 - P([A^c \cap B^c]^c) \quad \text{by properties of prob.} \\ &= 1 - P(A \cup B) \quad \text{by DeMorgan's Law} \\ &= 1 - 0.6 \quad \text{from part a)} \end{aligned}$$

$$P(A^c \cap B^c) = 0.4$$

$$\begin{aligned} c) \quad P(A \cup B \cup C) &= P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C) \\ &\quad \text{by inclusion-exclusion principle} \\ &= 0.3 + 0.4 + 0.5 - 0.1 - 0.2 - 0.3 + 0.01 \end{aligned}$$

$$P(A \cup B \cup C) = 0.61$$

$$\begin{aligned} d) \quad P(A^c \cap B^c \cap C^c) &= 1 - P([A^c \cap B^c \cap C^c]^c) \quad \text{by properties of prob.} \\ &= 1 - P(A \cup B \cup C) \quad \text{by DeMorgan's} \end{aligned}$$

$$P(A^c \cap B^c \cap C^c) = 1 - 0.61 = 0.39 \quad \text{from c)}$$

$$e) P((A^c \cap B^c) \cup C) = P(A^c \cap B^c) + P(C) - P((A^c \cap B^c) \cap C) \quad \text{by inclusion-exclusion}$$

$$\begin{aligned} P((A^c \cap B^c) \cap C) &= P(C) - P(A \cap C) - P(B \cap C) \\ &\quad + P(A \cap B \cap C) \\ P((A^c \cap B^c) \cap C) &= 0.5 - 0.2 - 0.3 + 0.01 \\ &= 0.01 \end{aligned}$$

$$= \underline{P(A^c \cap B^c)} + 0.5 - 0.01$$

$$= 0.4 \quad \text{from b)}$$

$$\boxed{P((A^c \cap B^c) \cup C) = 0.89}$$

2. (25 points, 5 points for each part)

Consider tossing a fair coin, and let  $X$  be the toss on which the first head appears. Let  $Y = \min\{X, 4\}$ . That is,  $Y = X$  if  $X = 1, 2, 3$ , and  $Y = 4$  otherwise.

- (a) Find the pmf of  $X$ .
- (b) Find the pmf of  $Y$ ,
- (c) Sketch the cdf of  $Y$ .
- (d) Let  $A$  be the event that  $Y$  is odd,  $B$  be the event that  $Y$  is even. Are  $A$  and  $B$  independent events? Justify your answer.
- (e) Find  $\mathbb{E}[Y]$ .

a) Each coin flip is independent, and equals 1 (resp. 0) with probability  $\frac{1}{2}$  (resp.  $\frac{1}{2}$ ).

$$\begin{aligned} P(X = k) &= P(\{k-1 \text{ Tails, then 1 Head}\}) \\ &= \underbrace{P(\{1st T\})}_{= 1/2} \cdot \underbrace{P(\{2nd T\})}_{= 1/2} \cdot \dots \cdot \underbrace{P(\{k-1st T\})}_{= 1/2} \cdot \underbrace{P(\{kth H\})}_{= 1/2} \end{aligned}$$

$$\boxed{P(X = k) = \frac{1}{2}^k, \text{ for } k = 1, 2, \dots}$$

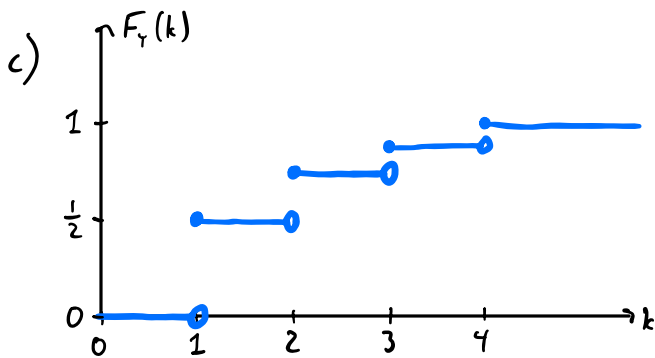
$$b) P(Y=k) = P(X=k) \text{ for } k=1, 2, 3$$

If  $X \geq 4$ , then  $Y=4$

$$P(Y=4) = P(X \geq 4) = 1 - P(X=1) - P(X=2) - P(X=3) \\ = 1 - \frac{1}{2} - \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^3$$

$$P(Y=4) = 0.125$$

$$P_Y(k) = \begin{cases} \frac{1}{2}^k & \text{for } k=1, 2, 3 \\ 0.125 & \text{for } k=4 \end{cases}$$



$$d) A = \{Y \text{ is odd}\} = \{Y=1, 3\}$$

$$B = \{Y \text{ is even}\} = \{Y=2, 4\}$$

$$A \cap B = \emptyset \Rightarrow P(A \cap B) = 0$$

$$\text{Since } P(A) > 0 \text{ and } P(B) > 0, \quad P(A)P(B) > 0$$

$\therefore A$  and  $B$  are not independent events

$$e) E[Y] = \sum_{k=1}^4 P_Y(k) \cdot k \\ = \left(\frac{1}{2}\right) \cdot 1 + \left(\frac{1}{4}\right) \cdot 2 + \left(\frac{1}{8}\right) \cdot 3 + \left(\frac{1}{8}\right) \cdot 4$$

$$E[Y] = 1.875$$

3. (20 points, 5 points for each part)

A diagnostic test for a disease has a probability 0.9 of giving a positive result when applied to an individual who has the disease, and a probability 0.01 of giving a (false) positive when applied to someone who does not have the disease. It is estimated that 0.1% of the population have the disease. Suppose we apply the test to a random person. Calculate the following probabilities:

- (a) The test result will be positive.
  - (b) Given a positive result, the individual tested has the disease.
  - (c) Given a negative result, the person does not have the disease.
  - (d) The probability that the person will be misclassified (i.e. the result of the test does not correspond to their state of disease).
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$$\begin{aligned} a) \quad P(\text{Test} +) &= P(\text{Test} + | \text{Disease} +) P(\text{Disease} +) + P(\text{Test} + | \text{Disease} -) P(\text{Disease} -) \quad \text{by Law of Total Prob.} \\ &= 0.9 \cdot 0.001 + 0.01 \cdot (1 - 0.001) \end{aligned}$$

$$P(\text{Test} +) = 1.09\%$$

$$\begin{aligned} b) \quad P(\text{Disease} + | \text{Test} +) &= \frac{P(\text{Test} + | \text{Disease} +) P(\text{Disease} +)}{P(\text{Test} +)} \quad \text{by Bayes' Rule} \\ &= \frac{90\% \cdot 0.1\%}{1.09\%} \end{aligned}$$

$$P(\text{Disease} + | \text{Test} +) = 8.3\%$$

$$\begin{aligned} c) \quad P(\text{Disease} - | \text{Test} -) &= \frac{P(\text{Test} - | \text{Disease} -) \cdot P(\text{Disease} -)}{P(\text{Test} -)} \quad \text{by Bayes' Rule} \\ &= \frac{99\% \cdot (1 - 0.001)}{1 - 1.09\%} \end{aligned}$$

$$P(\text{Disease} - | \text{Test} -) = 99.99\%$$

$$\begin{aligned}
 d) P(\text{Misclassified}) &= P(\text{Disease} +, \text{Test} -) + P(\text{Disease} -, \text{Test} +) \quad \text{by countable additivity} \\
 &= P(T-|D+) P(D+) + P(T+|D-) \cdot P(D-) \quad \text{by def. of conditional prob.} \\
 &= 0.1 \cdot 0.001 + 0.01 \cdot (1 - 0.001)
 \end{aligned}$$

$$P(\text{Misclassified}) = 1.01 \%$$

#### 4. (30 points, 5 points for each part)

Consider the random variables  $X, Y$  with the following joint PDF:

$$f(x, y) = \begin{cases} cxy & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- What value of  $c$  ensures that  $f(x, y)$  is a valid joint PDF.
- Find the marginal PDF of  $X$ .
- Find  $\mathbb{E}[X]$ .
- Find the conditional density of  $Y$  given that  $X = 0.5$ , i.e. find  $f_{Y|X}(y|X = 0.5)$ .
- Find  $\mathbb{E}[XY]$ .
- Are  $X$  and  $Y$  independent? Justify your answer.

a) Need:

$$\begin{aligned}
 1 &= \int_0^1 \int_0^1 cxy \, dx \, dy \\
 &= c \cdot \left( \int_0^1 x \, dx \right) \left( \int_0^1 y \, dy \right) \quad \text{pulling out constants} \\
 &= c \cdot \left[ \frac{1}{2} x^2 \right]_0^1 \left[ \frac{1}{2} y^2 \right]_0^1 \\
 1 &= c/4 \\
 c &= 4
 \end{aligned}$$

$$b) f_x(x) = \int_0^1 4 \cdot xy \, dy$$

$$= 4x \cdot \left[ \frac{1}{2} y^2 \right]_0^1$$

$$f_x(x) = 2x$$

$$c) E[X] = \int_0^1 x \cdot (2x) \, dx$$

$$= 2 \left[ \frac{1}{3} x^3 \right]_0^1$$

$$E[X] = \frac{2}{3}$$

$$d) f_{Y|X}(y | X = 1/2) = \frac{f(1/2, y)}{f_x(1/2)}$$

$$= \frac{4 \cdot \frac{1}{2} \cdot y}{2 \cdot \frac{1}{2}}$$

$$f_{Y|X}(y | X = 1/2) = 2y$$

$$e) E[XY] = \int_0^1 \int_0^1 xy \cdot 4xy \, dx \, dy$$

$$= 4 \cdot \left( \int_0^1 x^2 \, dx \right) \left( \int_0^1 y^2 \, dy \right) \quad \text{pulling out constants}$$

$$= 4 \cdot \left[ \frac{1}{3} x^3 \right]_0^1 \cdot \left[ \frac{1}{3} y^3 \right]_0^1$$

$$E[XY] = 4/9$$

$$f) \text{ Marginal of } Y: f_Y(y) = \int_0^1 4 \cdot xy \, dx$$

$$f_Y(y) = 2y \quad (\text{same as } b)$$

$$\text{Since: } f(x, y) = 4xy = 2x \cdot 2y = f_X(x) \cdot f_Y(y),$$

then  $X$  and  $Y$  are independent.